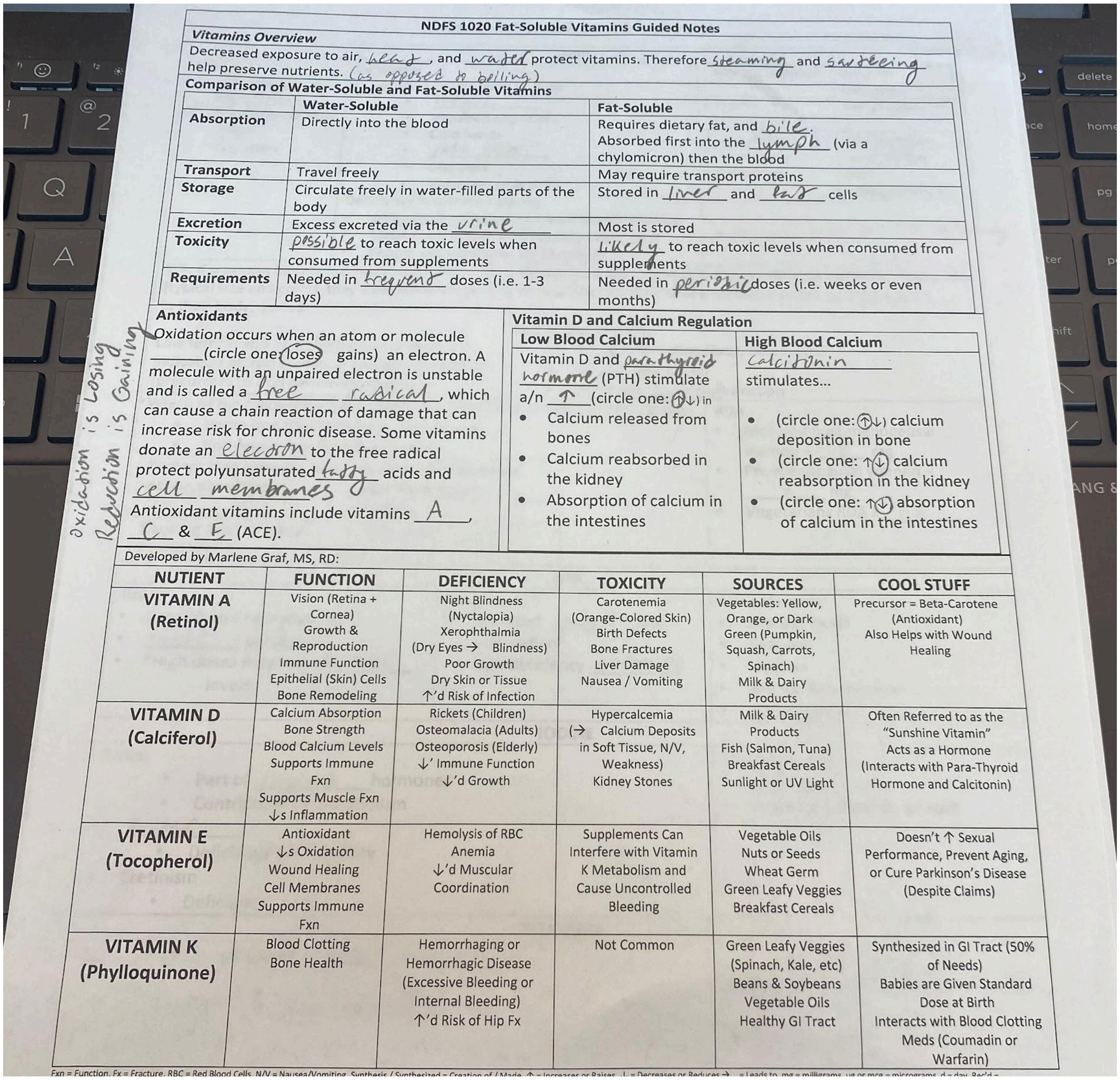


NDFS 1020 Fat-Soluble Vitamins Guided Notes

Vitamins Overview

Decreased exposure to air, heat, and water protect vitamins. Therefore steaming and sauteeing help preserve nutrients. (as opposed to boiling)

Comparison of Water-Soluble and Fat-Soluble Vitamins

	Water-Soluble	Fat-Soluble
Absorption	Directly into the blood	Requires dietary fat, and bile. Absorbed first into the lymph (via a chylomicron) then the blood
Transport	Travel freely	May require transport proteins
Storage	Circulate freely in water-filled parts of the body	Stored in liver and fat cells
Excretion	Excess excreted via the urine	Most is stored
Toxicity	possible to reach toxic levels when consumed from supplements	Likely to reach toxic levels when consumed from supplements
Requirements	Needed in frequent doses (i.e. 1-3 days)	Needed in periodic doses (i.e. weeks or even months)

Antioxidants

Oxidation occurs when an atom or molecule ______(circle one: loses gains) an electron. A molecule with an unpaired electron is unstable and is called a free radical, which can cause a chain reaction of damage that can increase risk for chronic disease. Some vitamins donate an electron to the free radical protect polyunsaturated fatty acids and cell membranes

Antioxidant vitamins include vitamins A, C & E (ACE).

Oxidation is Losing
Reduction is Gaining

Vitamin D and Calcium Regulation

Low Blood Calcium	High Blood Calcium
Vitamin D and parathyroid hormone (PTH) stimulate a/n ↑ (circle one: ↑↓) in • Calcium released from bones • Calcium reabsorbed in the kidney • Absorption of calcium in the intestines	Calcitonin stimulates... • (circle one: ↑↓) calcium deposition in bone • (circle one: ↑↓) calcium reabsorption in the kidney • (circle one: ↑↓) absorption of calcium in the intestines

Developed by Marlene Graf, MS, RD:

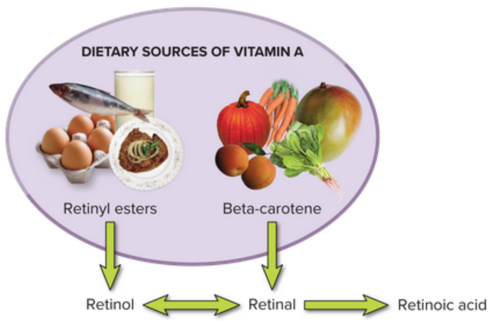
NUTRIENT	FUNCTION	DEFICIENCY	TOXICITY	SOURCES	COOL STUFF
VITAMIN A (Retinol)	Vision (Retina + Cornea) Growth & Reproduction Immune Function Epithelial (Skin) Cells Bone Remodeling	Night Blindness (Nyctalopia) Xerophthalmia (Dry Eyes → Blindness) Poor Growth Dry Skin or Tissue ↑'d Risk of Infection	Carotenemia (Orange-Colored Skin) Birth Defects Bone Fractures Liver Damage Nausea / Vomiting	Vegetables: Yellow, Orange, or Dark Green (Pumpkin, Squash, Carrots, Spinach) Milk & Dairy Products	Precursor = Beta-Carotene (Antioxidant) Also Helps with Wound Healing
VITAMIN D (Calciferol)	Calcium Absorption Bone Strength Blood Calcium Levels Supports Immune Fxn Supports Muscle Fxn ↓s Inflammation	Rickets (Children) Osteomalacia (Adults) Osteoporosis (Elderly) ↓' Immune Function ↓'d Growth	Hypercalcemia (→ Calcium Deposits in Soft Tissue, N/V, Weakness) Kidney Stones	Milk & Dairy Products Fish (Salmon, Tuna) Breakfast Cereals Sunlight or UV Light	Often Referred to as the "Sunshine Vitamin" Acts as a Hormone (Interacts with Para-Thyroid Hormone and Calcitonin)
VITAMIN E (Tocopherol)	Antioxidant ↓s Oxidation Wound Healing Cell Membranes Supports Immune Fxn	Hemolysis of RBC Anemia ↓'d Muscular Coordination	Supplements Can Interfere with Vitamin K Metabolism and Cause Uncontrolled Bleeding	Vegetable Oils Nuts or Seeds Wheat Germ Green Leafy Veggies Breakfast Cereals	Doesn't ↑ Sexual Performance, Prevent Aging, or Cure Parkinson's Disease (Despite Claims)
VITAMIN K (Phylloquinone)	Blood Clotting Bone Health	Hemorrhaging or Hemorrhagic Disease (Excessive Bleeding or Internal Bleeding) ↑'d Risk of Hip Fx	Not Common	Green Leafy Veggies (Spinach, Kale, etc) Beans & Soybeans Vegetable Oils Healthy GI Tract	Synthesized in GI Tract (50% of Needs) Babies are Given Standard Dose at Birth Interacts with Blood Clotting Meds (Coumadin or Warfarin)

Fxn = Function, Fx = Fracture, RBC = Red Blood Cells, N/V = Nausea/Vomiting, Synthesis / Synthesized = Creation of / Made, ↑ = Increases or Raises, ↓ = Decreases or Reduces → = Leads to, mg = milligrams, ug or mcg = micrograms, d = day, Rec'd =

Know

Vitamin A

- Functions
    - Cell production, growth and development, function, and maintenance
    - Fertility
    - Production and activity of white blood cells
        - Immune system by extension
    - Bone growth and development

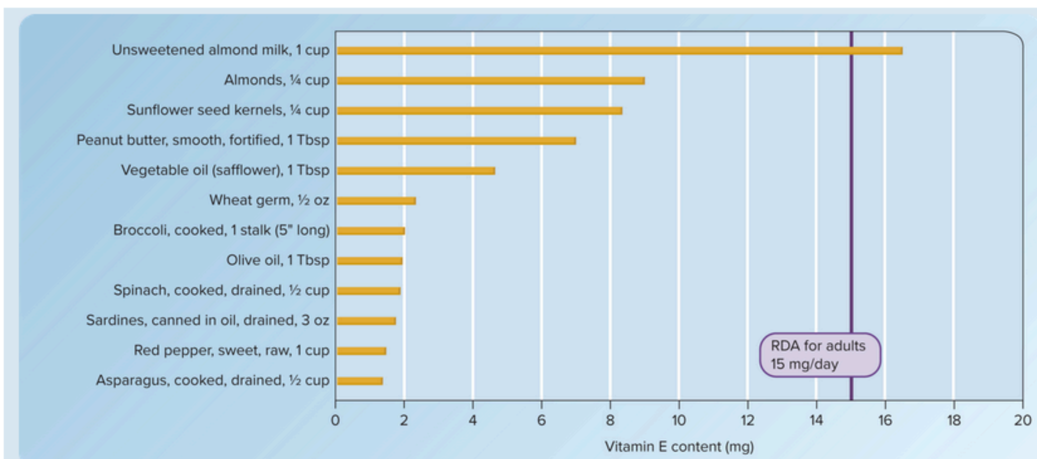
- Vision
    - Especially night vision
- Food sources
    - Preformed vit A: Liver, butter, fish liver oils, eggs, fortified cow's milk/yogurt/margarine/cereals
    - Beta carotene: carrots, spinach, leafy greens, pumpkins, sweet potatoes, broccoli, mangoes, cantaloupe
    - 

    - Body retains 1 microgram of retinol from every 12 micrograms of beta carotene in food
    - Vitamin A precursors in plants are not absorbed as well as retinol in animal foods
    - 90% of vitamin A is stored in the liver
- Dietary adequacy (deficiency and toxicity)
    - RDA = 700-900 micrograms
    - Generally met in standard diets in US, but people of concern:
        - Kids who don't eat enough veggies
        - Ppl with very low fat diets
        - Low income urban residents
        - Older adults
        - Alcoholics
        - Ppl with fat malabsorption
        - Ppl with liver diseases
    - Deficiency
        - Rough and bumpy skin
        - Buildup of keratin in the cornea and mucus producing cells in the eye (dry eye/xerophthalmia)
        - Negative impact on immune system leads to increased risk of infection
        - Impacts on fetal eye health, organ and skeletal development (NOTE: excess vitamin A during pregnancy is a teratogen and can cause birth defects and spontaneous abortion)
    - Toxicity
        - UL = 3000 micrograms a day
        - Liver damage
        - Headache, nausea, vomiting, visual disturbances, hair loss, bone pain, bone fractures
        - carotenemia/yellow skin (usually harmless)

Vitamin E

- Functions
    - Antioxidant, protects PUFA in cell membranes from being damaged by free radicals
        - Helps prevent atherosclerosis (causes heart attack/stroke, cancer, premature cellular aging and death)
    - Maintain nervous tissue
    - Immune system function
- Food sources
    - Sunflower seeds, almonds, almond milk, peanuts, plant oils (especially sunflower, safflower, canola, and olive) and products made from them (margarine, salad dressings), fish, whole grains, nuts, seeds, some veggies

      TABLE 9.7 Vitamin E Content of Selected Foods

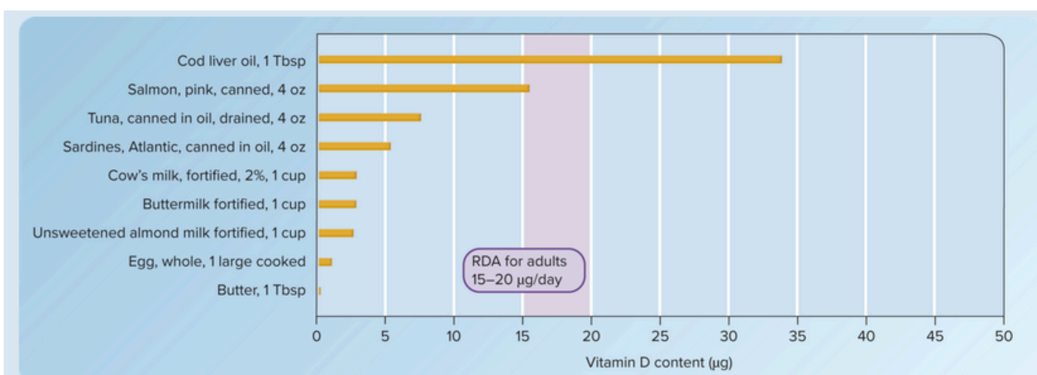
    -
    - Most vit E in grains is lost during milling. Also sensitive to oxygen exposure, metals, light, high temps like frying
- Dietary adequacy (deficiency and toxicity)
    - RDA: 15 mg a day
    - Most americans don't meet this, but deficiency is rare because the body stores it in fat for long periods of time
    - Deficiency:
        - Damage to nervous system/nerve damage
        - Loss of neuromuscular control
        - Blindness
        - Reduced immune function and a form of anemia in infants
        - Ppl with fat absorption issues are at risk
    - Toxicity
        - UL: 1000 mg a day
        - Interferes with vit K's role in blood clotting and can lead to hemorrhages

        - Ppl on blood thinners should ask a doctor before taking supplements

Vitamin D

- Functions
    - Calcium regulation and absorption
        - Low blood calcium prompts vit D to work with parathyroid hormone (PTH) to signal bones to release calcium
        - PTH also signals kidneys to increase vit D production and decrease elimination of calcium in urine
    - maintaining phosphorus levels in blood and absorption
    - Production and maintenance of bones
    - Regulate neuromuscular and immune function
    - Reduce inflammation
    - *May* reduce risk for heart disease, cancer, diabetes mellitus, multiple sclerosis, asthma, and depression
- Food sources
    - Fish liver oils and fatty fish like salmon/herring/catfish
    - Fortified Cows milk, cereals, orange juice, etc

TABLE 9.6 Vitamin D Content of Selected Foods



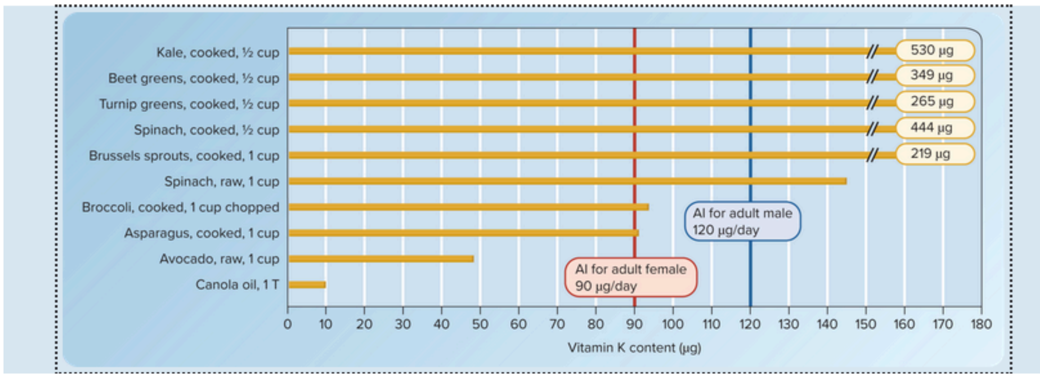
    -
    - Also sunlight :)
- Dietary adequacy (deficiency and toxicity)
    - RDA = 15 micrograms a day
    - Most Americans don't meet this
    - more at risk to be deficient:
        - Darker skintones
        - Overweight
        - Indoors most of the day
        - Use electronics multiple hours a day
        - Don't frequently drink cows milk
        - Ppl living north of 33rd parallel, especially in winter
    - Deficiency:

        - Rickets (kids should have at least 400 IU a day through adolescence)
        - Osteomalacia (adult form of rickets, low levels of calcium in bones make them soft and prone to bending or breaking. Also muscle weakness)
        - At risk:
            - Adults confined indoors most of the day
            - Ppl who are fully covered when going outside
            - Ppl with kidney, liver, or intestinal diseases
            - Older adults (decreased rates of conversion from previtamin to active forms of vit D)
    - Toxicity
        - UL: 100 micrograms a day for adults (4000 IU)
        - Calcium deposits in soft tissues like kidneys/heart/blood vessels, causing impaired cell function and cell death
        - Muscular weakness
        - Loss of appetite
        - Diarrhea
        - Vomiting
        - Mental confusion

Vitamin K

- Functions
    - Blood clotting
    - Bone formation
    - Inhibition of bone breakdown
    - Regulation of calcium levels in body, prevents deposition in soft tissues
        - Prevents arterial calcification
    - *May* promote insulin sensitivity, suppress cancer progression, and improve cognitive performance
- Food sources
    - Leafy greens like kale, turnip greens, salad greens, cabbage, spinach, broccoli, green beans, egg yolks, butter, some cheeses, liver, beef, some fermented foods, soybean and canola oils and their products

TABLE 9.8 Vitamin K Content of Selected Foods



-
- Dietary adequacy (deficiency and toxicity)
    - No RDA, but AI is 120 micrograms a day for males and 90 micrograms a day for females
    - Deficiency
        - Rare in adults, but ppl at risk:
            - Taken long courses of antibiotics
            - Very low fat diets
            - Liver diseases
            - Conditions that impair fat absorption, like cystic fibrosis
        - Increased time for blood to clot
        - Blood vessel calcification
        - Increased risk of bone fracture
        - Common in most infants, so an injection is standard immediately after birth to prevent issues with blood clotting
    - Toxicity
        - No UL, not harmful

Skim

Intro to vitamins

- First discovered in 1912 with thiamin, but by end of the 1900s they had added riboflavin, niacin, b6, b12, pantothenic acid, folate, ascorbic acid, A, D, E, and K
- Probably all of them, if any were left undiscovered then ppl surviving off of formulas would not be able to survive as long as they do
- Megadose = way more than recommended

Vitamins: the basics

- Vitamin = complex organic compound that regulates metabolic practices and that is:
    - An essential nutrient (body can't make any/enough of it to maintain good health)

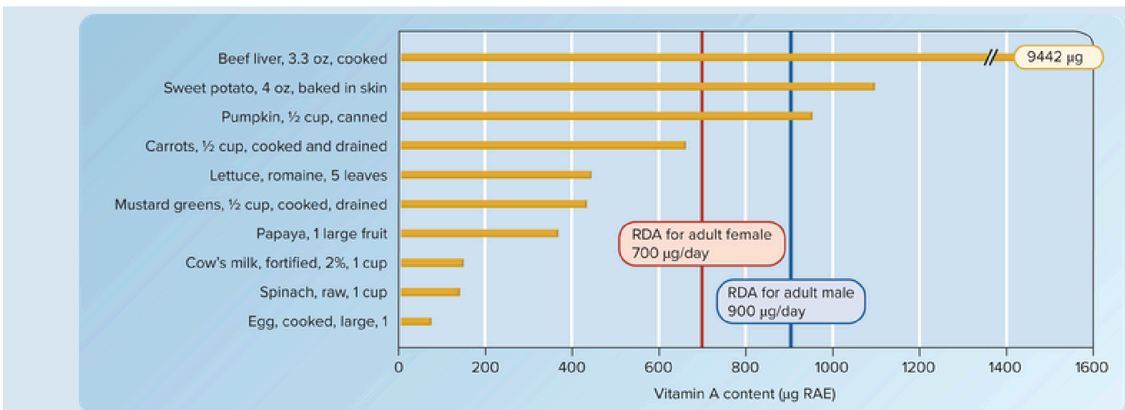
    - Occurs naturally in commonly eaten foods
    - Symptoms of deficiency start to appear when it is not present in diet or can't be metabolized properly
    - Deficiency symptoms go away if the missing substance is supplied
- Can be fat soluble
    - A
    - D
    - E
    - K
- Or water soluble
    - Thiamin
    - Riboflavin
    - Niacin
    - B6
    - B 12
    - Pantothenic acid
    - Folate
    - Biotin
    - C

- Antioxidants give up electrons to free radicals to help them become stable (so they don't oxidize PUFA in cell membranes or DNA in the nucleus)
-

Vitamin A - carotenoids

- Types
    - Beta carotene
    - Lutein
    - Zeaxanthin
    - Lycopene
- Sources
    - Leafy greens
    - Tomato products
    - Darkly pigmented fruits and veggies
- Body can’t convert all to vit. A, but they are helpful antioxidants

TABLE 9.4 Vitamin A Content of Selected Foods

- 

Vitamin A and chronic disease

- Higher levels in diet = lower risk of certain cancers, heart disease, and age related macular degeneration (blindness)
    - Due to higher levels of beta carotene and other antioxidant carotenoids
- Results
    - Vitamin A/ beta carotene intake and cancer/heart disease = mixed
    - Age related macular degeneration = mixed. If you already have it they can be helpful, but prob don't help prevent it entirely. Also long term supplement use is bad

Vitamin E and chronic disease

- Previously thought to lower risk of heart disease, cancer (particularly lung in smokers), might help slow Alzheimer’s. Not as sure now, and definitely increases risk of hemorrhagic stroke

Vitamin K and osteoporosis

- Important for maintaining bone strength
- More research is needed for dosing effects and safety of vit K for bone health